UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

School of ICT

Department of Computer and Software Engineering

WEB DESIGN TECHNICAL REPORT

UBUZIMA HUB

An Interactive Youth Health Learning Platform

Submitted in Module Web Design

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ABSTRACT

Many young people still lack access to reliable, engaging, and interactive health education. Misinformation regarding HIV/AIDS, mental health, sexual and reproductive health, menstrual health, sexually transmitted diseases (STDs), and non-communicable diseases (NCDs) continues to affect communities, especially among the youth.

This project presents UBUZIMA HUB, a web-based learning platform that provides accurate, accessible, and interactive health education. Users can create accounts, enroll in health courses, study structured lessons, participate in quizzes, and earn awards based on their performance, while administrators manage the educational content. The system was developed using HTML, CSS, and JavaScript, is version-controlled with Git, and is deployed publicly through GitHub Pages, providing a responsive interface suitable for learners on any device.

The implementation of UBUZIMA HUB demonstrates how web technologies can improve youth access to trustworthy health information while promoting continuous learning through interactive lessons, quizzes, and educational resources.

Keywords: *Web Design, Health Education, Learning Platform, Youth Empowerment, HTML, CSS, JavaScript, UBUZIMA HUB.*

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
UI	User Interface
ERD	Entity Relationship Diagram
DBMS	Database Management System
URL	Uniform Resource Locator
HIV	Human Immunodeficiency Virus
AIDS	Acquired Immunodeficiency Syndrome
STD	Sexually Transmitted Disease
NCD	Non-Communicable Disease
WHO	World Health Organization
RBC	Rwanda Biomedical Centre
ICT	Information and Communication Technology

CHAPTER ONE: INTRODUCTION

1.1 Historical Background

Health education has always been an essential component of public health. Traditionally, health information in Rwanda and across Africa has been delivered through community health workers, schools, radio programmes, printed brochures, and outreach campaigns. While these methods have played an important role, they face significant limitations: printed materials become outdated quickly, radio broadcasts are one-directional, and face-to-face sessions can only reach a limited audience at a given time. The emergence of the Internet and the rapid penetration of smartphones have created new opportunities: organizations such as the World Health Organization (WHO) and UNAIDS now publish health information online, and in Rwanda, initiatives driven by the Ministry of Health and the Rwanda Biomedical Centre (RBC) have promoted digital health services while national ICT programs have expanded internet access.

Despite this progress, most reliable health content available online is written for international audiences, is scattered across many websites, and is rarely presented in an interactive, youth-friendly format. Young people frequently turn to social media for answers to sensitive health questions, where misinformation spreads easily. There is therefore a clear need for a dedicated, interactive, and trustworthy platform that consolidates youth health education in one place. UBUZIMA HUB — from the Kinyarwanda word "ubuzima", meaning "health" or "life" — was conceived to respond to this need.

1.2 Problem Statement

Many young people lack access to accurate, engaging, and age-appropriate health information. Existing sources are often fragmented, overly technical, outdated, or presented in static formats that do not encourage active learning. As a result, myths and misinformation about HIV/AIDS, mental health, sexual and reproductive health, menstrual health, STDs, and non-communicable diseases persist among the youth, leading to stigma, risky behaviour, late testing, and poor health decisions. There is currently no single, interactive, youth-focused web platform that combines structured lessons, self-assessment quizzes, and motivational challenges on these health topics. UBUZIMA HUB addresses this gap by providing a free, responsive, and interactive web-based learning platform.

1.3 Objectives

1.3.1 General Objective

To design and develop an interactive web-based health learning platform (UBUZIMA HUB) that provides young people with accurate, accessible, and engaging health education.

1.3.2 Specific Objectives

- To design a responsive, user-friendly website using HTML, CSS, and JavaScript.
- To organise reliable health content into structured learning modules covering HIV/AIDS, mental health, sexual and reproductive health, menstrual health, STDs, and non-communicable diseases.

- To develop interactive lessons that allow learners to expand, collapse, and navigate topics without reloading pages.
- To implement quizzes that assess learners' understanding and provide instant feedback.
- To create a challenges section that motivates learners through practical health-related activities.
- To provide account (registration/login) interfaces that prepare the system for future personalisation features.
- To deploy the website publicly using Git, GitHub, and GitHub Pages.

1.4 Expected Outcomes

- A fully functional, publicly accessible health education website.
- Improved access to trustworthy health information for young people.
- Increased health knowledge and awareness among users, verified through quizzes.
- Reduced stigma and misinformation around sensitive health topics.
- A reusable foundation that can later be extended with a database, backend, and mobile application.

1.5 Project Rationale

Health literacy is a key determinant of individual and community well-being. Rwanda's young population is increasingly connected to the Internet, making a web platform an efficient, low-cost channel for health education. Building the platform with standard web technologies (HTML, CSS, and JavaScript) ensures that it is lightweight, fast to load even on low-bandwidth connections, easy to maintain, and free to host on GitHub Pages. Academically, the project applies the concepts taught in the Web Design module — page structure, styling, responsiveness, interactivity, and deployment — to a socially meaningful real-world problem.

1.6 Scope of the Project

The project covers the design, implementation, testing, and deployment of a front-end web application consisting of the Home, About, Learn (with six health modules), Quiz, Challenges, Account, and Contact pages. The current version is a static website: user accounts, quiz scores, and challenges are handled on the client side, and permanent data storage through a database is proposed as future work.

1.7 Limitations

- No backend server or database integration in the current version; data is not permanently stored.
- Quiz scores and learning progress are not saved between sessions.
- Content updates require manual editing of the source files by the administrator.
- The platform currently supports English only.

CHAPTER TWO: SYSTEM ANALYSIS AND DESIGN

2.1 System Analysis

2.1.1 Analysis of the Existing Situation

At present, young people seeking health information rely on scattered sources: search engines, social media, printed brochures, and occasional school or community sessions. These sources are either unverified (social media), static (brochures), or unavailable on demand (physical sessions). Existing international health websites are reliable but are not interactive, are not tailored to youth, and do not assess whether the reader has actually understood the content.

2.1.2 Proposed System

UBUZIMA HUB is proposed as an interactive web platform that consolidates reliable youth health content into structured modules, verifies understanding through quizzes, and sustains motivation through challenges. The system is accessible from any device with a web browser and requires no installation.

2.1.3 Functional Requirements

Table 2.1: Functional Requirements

ID	Requirement	Description
FR1	Account interfaces	Users can fill registration and login forms with client-side validation.
FR2	Browse topics	Users can browse six health learning modules from the Learn page.
FR3	Interactive lessons	Users can expand/collapse lessons and switch topics without page reload.
FR4	Quizzes	Users can answer multiple-choice questions and receive instant scores and feedback.
FR5	Challenges	Users can view and participate in health challenges published by the administrator.
FR6	Contact	Users can view contact details and submit inquiries through a contact form.
FR7	Content management	The administrator can update lessons, quizzes, categories, and challenges.

2.1.4 Non-Functional Requirements

Table 2.2: Non-Functional Requirements

ID	Requirement	Description
NFR1	Usability	The interface must be simple, consistent, and understandable to first-time users.
NFR2	Responsiveness	The layout must adapt to desktop, tablet, and mobile screen sizes.
NFR3	Performance	Pages must load quickly, even on low-bandwidth connections.
NFR4	Compatibility	The site must work on Chrome, Edge, and Firefox.
NFR5	Maintainability	Code must be organised so content can be updated easily.
NFR6	Accessibility	Text must be readable, with clear contrast and legible typography.

2.1.5 User Description

Table 2.3: User Description

User	Role in the System
Learner (Youth user)	Registers/logs in, browses health topics, studies lessons, takes quizzes, joins challenges, and contacts the administrator.
Administrator	Manages educational content, quizzes, categories, challenges, and awards; responds to user inquiries.
Visitor	Browses public pages (Home, About, Learn) before creating an account.

2.2 System Design

2.2.1 System Architecture

The system follows a three-layer static-web architecture. The client layer consists of users' browsers on desktops, tablets, and smartphones. The front-end layer is the UBUZIMA HUB website itself, built with HTML5 (structure), CSS3 (styling and responsive layout), and JavaScript (interactivity). The hosting layer uses Git for version control, GitHub for source hosting, and GitHub Pages for public deployment over HTTPS.

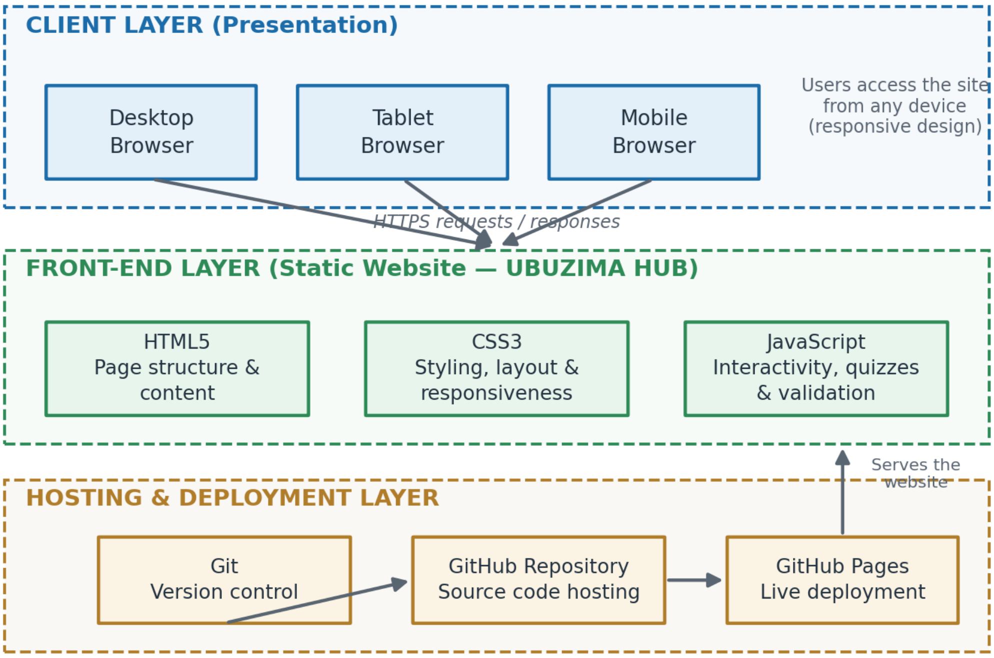

Figure 2.1: System Architecture of UBUZIMA HUB

2.2.2 Use Case Diagram

The use case diagram below shows how the two main actors interact with the system. The learner registers, logs in, browses topics, studies lessons, takes quizzes, participates in challenges, and contacts the administrator. The administrator logs in, manages content and quizzes, and publishes challenges.

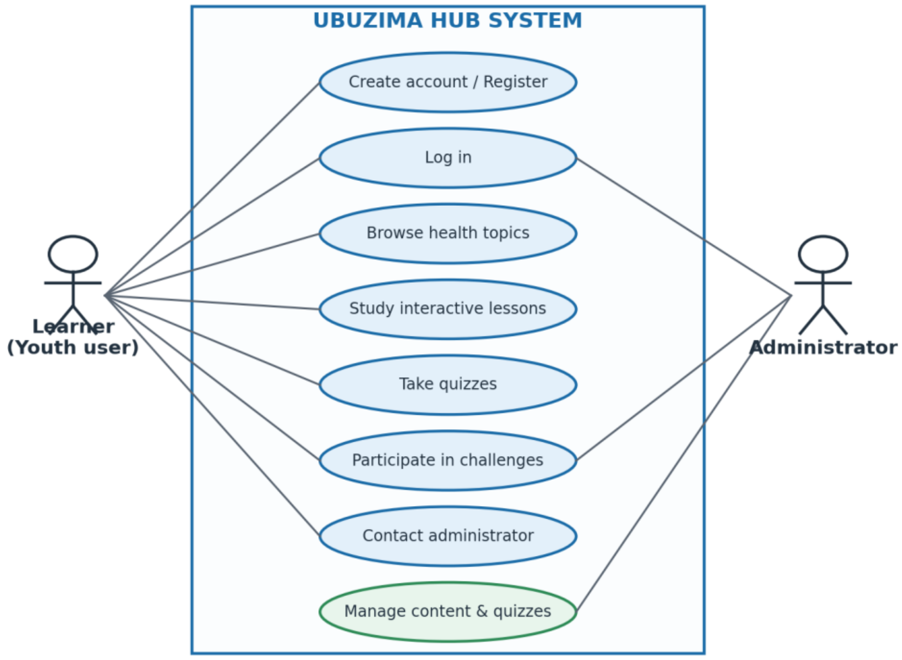

Figure 2.2: Use Case Diagram

2.2.3 Entity Relationship Diagram

Although the current version is a static website, the data model below was designed to guide future database integration. A user enrols in one or more topics; each topic contains lessons and quizzes; and completed quizzes generate awards for the user.

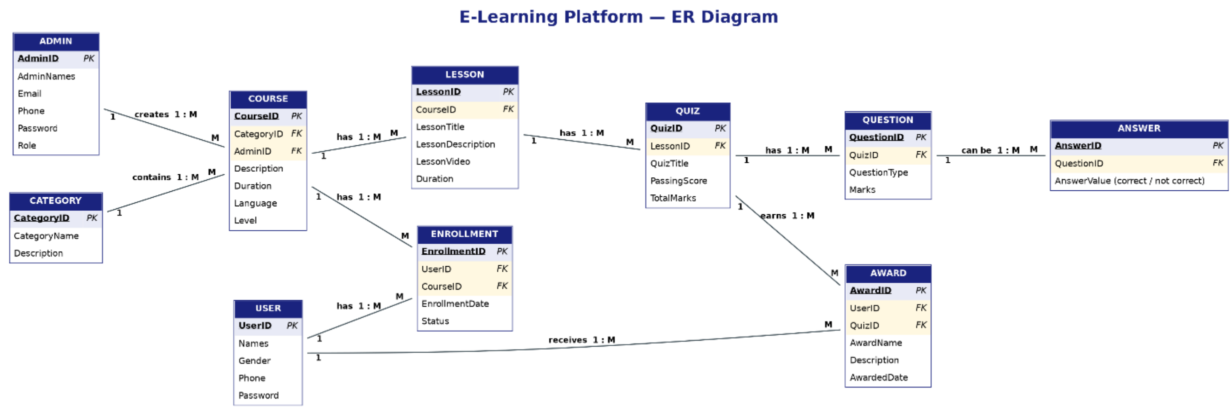

Figure 2.3: Entity Relationship Diagram (ERD)

Table 2.4: Proposed Database Tables

Table	Key Fields	Purpose
admin	AdminID (PK)	Stores administrator accounts (names, email, phone, password, role) who manage the platform and create courses.
category	CategoryID (PK)	Groups courses into categories with a name and description for easier browsing and organization.
course	CourseID (PK); CategoryID (FK); AdminID (FK)	Stores course details (description, duration, language, level) and links each course to its category and the admin who created it.

Table	Key Fields	Purpose
Lesson	LessonID (PK); CourseID (FK)	Holds the individual lessons of a course, including title, description, video content, and duration.
quizz	QuizID (PK); LessonID (FK)	Stores quizzes attached to lessons, with title, passing score, and total marks used for assessment.
question	QuestionID (PK); QuizID (FK)	Contains the questions belonging to each quiz, with question type and marks allocated.
answer	AnswerID (PK); QuestionID (FK)	Stores possible answers for each question and indicates whether each answer is correct or not correct.
user	UserID (PK)	Stores learner accounts (names, gender, phone, password) for people enrolling in courses.
enrollment	EnrollmentID (PK); UserID (FK); CourseID (FK)	Links users to the courses they join, recording the enrollment date and status (junction table between User and Course).
award	AwardID (PK); UserID (FK); QuizID (FK)	Records certificates/awards given to users after passing quizzes, with award name, description, and date awarded.

2.2.4 Navigation Structure

The navigation structure shows how pages are linked. The Home page provides access to all main pages through the navigation menu. The Learn page branches into the six health modules, each of which links to its topic quiz. Topic switching within the Learn page is handled by JavaScript without page reloads.

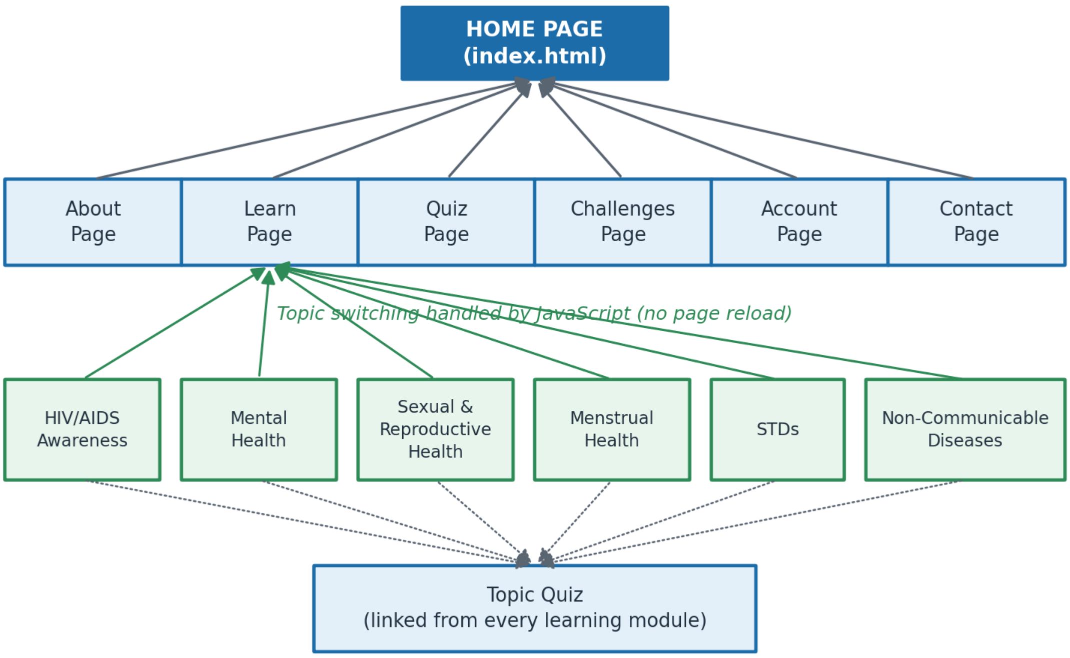

Figure 2.4: Website Navigation Structure

2.2.5 User Learning Flowchart

The flowchart below describes the typical learning journey: the user opens the website, registers or logs in, selects a health topic, studies the lessons, and takes the quiz. If the user does not pass, they are directed back to review the lessons; if they pass, an award is earned and the flow completes.

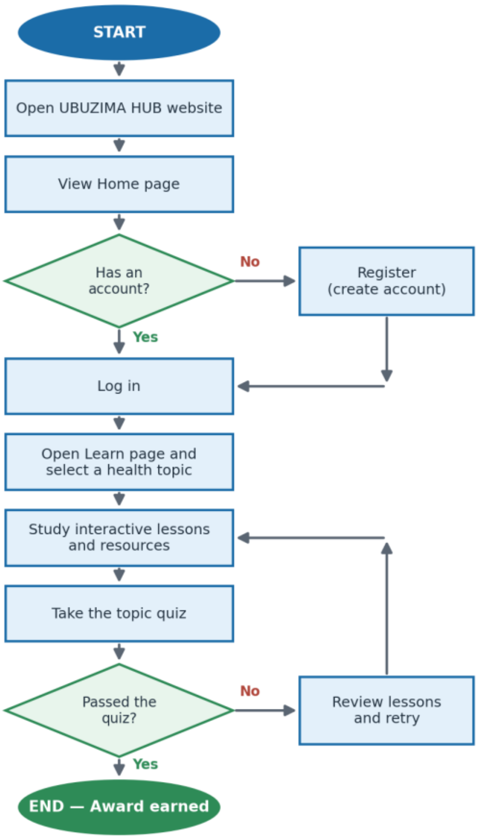

Figure 2.5: User Learning Process Flowchart

2.2.6 Software Requirements

Table 2.5: Software Requirements

Software	Purpose
Visual Studio Code	Writing and editing HTML, CSS, and JavaScript source code.
Google Chrome / Edge / Firefox	Testing, debugging, and browsing the website.
Git	Version control of the project source code.
GitHub & GitHub Pages	Hosting the repository and deploying the live website.
Windows 10/11 Operating System	Development environment.

CHAPTER THREE: SYSTEM IMPLEMENTATION

3.1 Introduction

This chapter presents the implementation of the UBUZIMA HUB learning platform. It explains the technologies used to develop the website, the implementation process, and the functionality of each webpage, together with screenshots of the implemented interfaces. The implementation focused on creating a responsive, user-friendly, and interactive educational website that enables young people to access reliable health information, participate in quizzes, and improve their knowledge in an engaging learning environment.

3.2 Development Environment

The UBUZIMA HUB website was developed using modern web development technologies, summarised in Table 3.1.

Table 3.1: Development Environment and Tools

Tool	Purpose
Visual Studio Code	Source code editor
HTML5	Website structure
CSS3	Styling and layout
JavaScript	Interactive functionalities
Git / GitHub / GitHub Pages	Version control, source hosting, and deployment
Google Chrome	Testing and debugging

3.3 Programming Languages Used

HTML was used to create the structure of every webpage — Home, About, Learn, Account, Quiz, Challenges, and Contact — defining headings, images, buttons, forms, navigation menus, and the educational content itself.

CSS was used to improve the appearance of the website: colours, layout, responsive design, navigation bar styling, buttons, cards, tables, image presentation, and footer design. The selected colour theme (blue and green) represents health, trust, education, and growth.

JavaScript provides interactivity throughout the website: topic switching on the Learn page, opening and closing lessons, quiz interactions, navigation behaviour, form validation, and user feedback — allowing users to interact with the content without reloading pages.

3.4 Home Page Implementation

The Home page is the first page displayed when visitors open the UBUZIMA HUB website. Its purpose is to introduce the platform and encourage visitors to begin learning. The page contains the website logo, navigation menu, hero section, learning categories, mission section, learning experience gallery, and footer, designed with attractive colours and organised content that motivates users to explore the platform.

Key features: *responsive navigation, call-to-action buttons, educational cards, learning images, mission statement, footer.*

3.5 About Page Implementation

The About page provides detailed information about UBUZIMA HUB. It explains the vision, mission, objectives, core values, why the platform was created, and the benefits it offers to young people. The page also introduces visitors to the importance of health education and encourages them to actively participate in the learning process.

Key features: *vision and mission sections, objectives, core values, Why Choose UBUZIMA HUB, call-to-action section.*

3.6 Learn Page Implementation

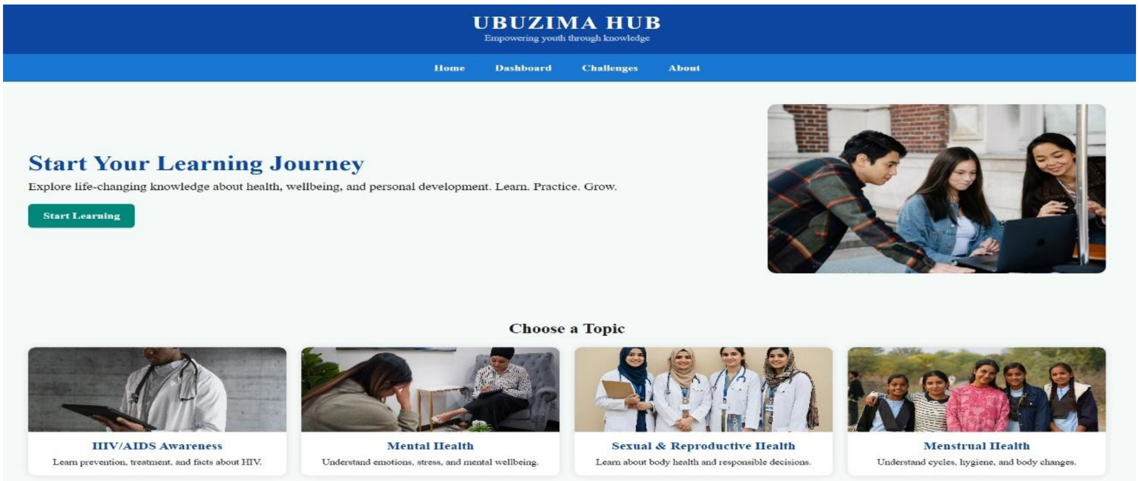

Figure 3.1: Learn Page

The Learn page is the heart of the UBUZIMA HUB platform. It organises educational materials into health-related categories, allowing learners to access reliable information in an interactive way. Each topic contains a hero image, topic description, statistics, lessons, key facts, learning resources, and a quiz link. Users can switch between topics without reloading the page through JavaScript. The topics included are: HIV/AIDS Awareness, Mental Health, Sexual and Reproductive Health, Menstrual Health, Sexually Transmitted Diseases, and Non-Communicable Diseases.

Key features: *interactive topic navigation, expandable lessons, educational images, resource links, statistics, responsive design.*

3.7 HIV/AIDS Learning Module

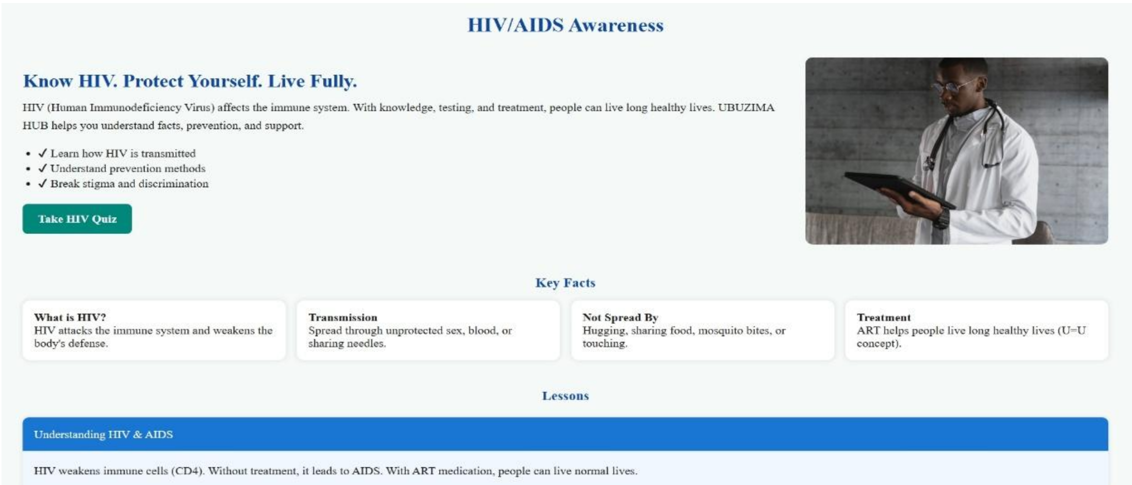

Figure 3.2: HIV/AIDS Awareness Lesson

This learning module teaches users about HIV/AIDS, including its definition, transmission, prevention, treatment, living positively, and fighting stigma. Interactive lessons allow users to expand or collapse information for easier reading.

Key features: *hero banner, facts cards, expandable lessons, learning resources, statistics, quiz button.*

3.8 Mental Health Module

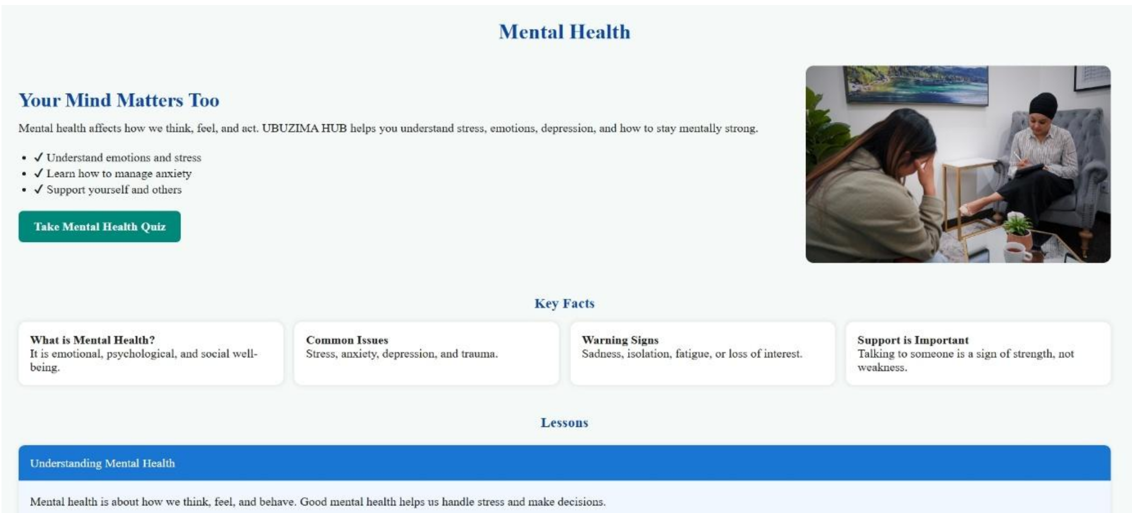

Figure 3.3: Mental Health Lesson

This section educates learners about mental well-being. Topics include stress, depression, anxiety, PTSD, emotional support, and self-care. The module encourages users to seek help and reduce stigma around mental health.

Key features: *educational images, expandable lessons, statistics, helpful resources, interactive learning cards.*

3.9 Menstrual Health Module

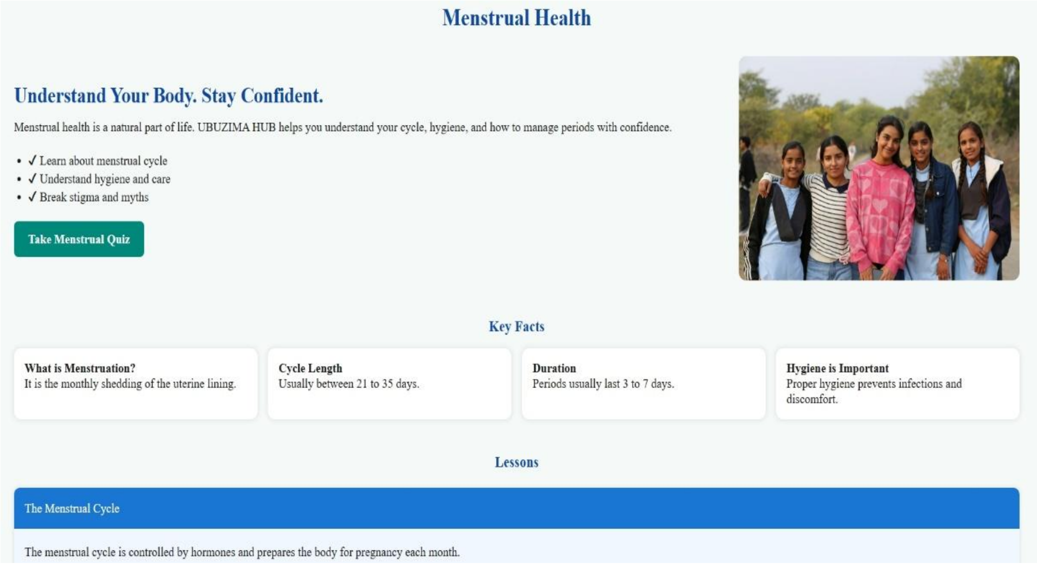

Figure 3.4: Menstrual Health Learning Module

The Menstrual Health module educates girls, boys, and the general community about menstruation and menstrual hygiene management. It provides scientifically accurate information while helping to reduce myths, misconceptions, and stigma, promoting confidence, dignity, and healthy practices. The main topics covered are: understanding menstruation, the menstrual cycle, menstrual hygiene, common myths and misconceptions, menstrual disorders, nutrition during menstruation, and school attendance during menstruation.

Key features: *interactive lessons, educational illustrations, resource links, key facts, quiz preparation, responsive design.*

3.10 Account Management Page

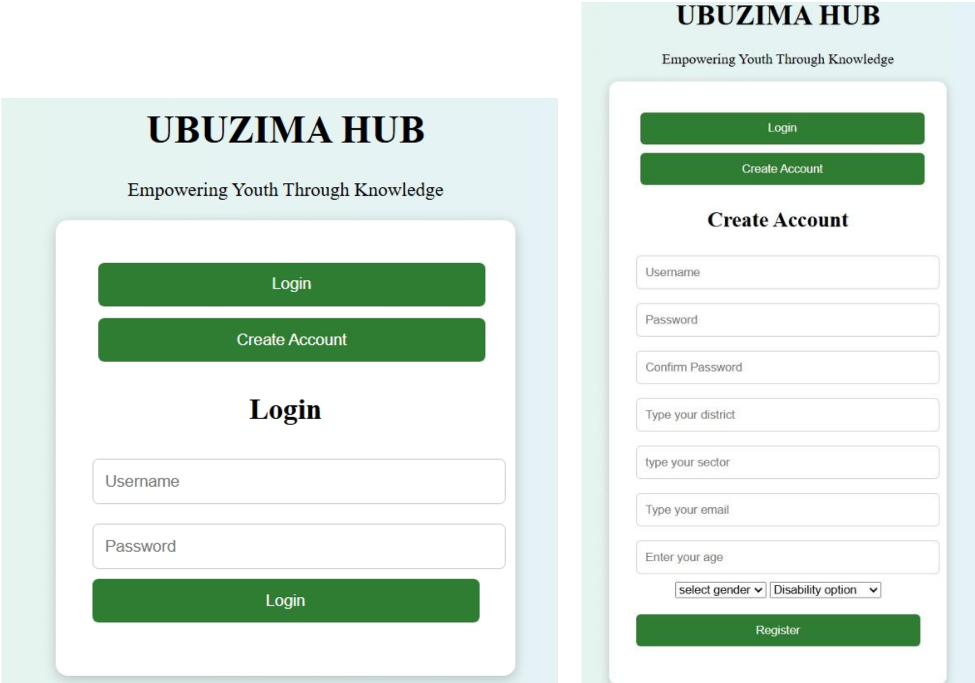

Figure 3.5: Account Page (Login and Registration)

The Account page enables users to register for a new account or log in to an existing account. Creating an account allows learners to access personalised educational content and prepares the system for future features such as progress tracking, quiz history, and rewards. Although the current version uses HTML, CSS, and JavaScript, the interface has been designed to integrate easily with a future database and backend system.

Key features: *user registration form, login form, password field, form validation, responsive interface, user-friendly design.*

3.11 Quiz Page

The Quiz page assesses learners' understanding after they complete lessons. Each quiz contains multiple-choice questions based on the educational content available in the Learn section. The page encourages active learning by allowing users to test their knowledge immediately after studying, increasing knowledge retention and motivating learners through interactive participation.

Key features: *multiple-choice questions, score calculation, instant feedback, user-friendly interface, responsive design.*

3.12 Challenges Page

The Challenges page presents special educational activities created by the administrator. Challenges encourage users to participate beyond normal lessons by completing additional tasks that improve practical knowledge and life skills — for example, the weekly health awareness challenge, mental wellness challenge, HIV awareness campaign challenge, healthy lifestyle challenge, and community engagement activities.

Key features: *list of active challenges, challenge descriptions, participation instructions, motivational design, future reward integration.*

3.13 Responsive Design

UBUZIMA HUB was designed using responsive web design principles. The interface automatically adjusts according to different screen sizes, allowing users to access learning materials from desktops, laptops, tablets, and smartphones. Responsive design improves accessibility and provides a consistent user experience across devices.

3.14 Webpages status

Table 3.2 Webpage status table

Webpages done	Webpages in progress
✓ Homepage ✓ Learning page* ✓ About page ✓ Account Page	• Quiz page • Challenges page • User dashboard and Admin dashboard • Rewards page

Key features: *flexible layout, responsive images, adaptive navigation, mobile-friendly buttons, readable typography.*

3.15 Summary

This chapter has presented the implementation of the UBUZIMA HUB learning platform: the development environment, the programming languages used, and the functionality of each major webpage. The implementation demonstrates how HTML, CSS, and JavaScript were used to build an interactive and user-friendly educational website focused on improving health knowledge among young people.

CHAPTER FOUR: CONCLUSION AND RECOMMENDATIONS

4.1 Introduction

This chapter summarises the achievements of the UBUZIMA HUB project, evaluates whether the objectives were met, and provides recommendations for future enhancements.

4.2 Conclusion

The development of UBUZIMA HUB successfully addressed the problem of limited access to reliable and engaging health education among young people. The platform provides educational content on HIV/AIDS awareness, mental health, sexual and reproductive health, menstrual health, sexually transmitted diseases, and non-communicable diseases through an interactive web interface.

The website was developed using HTML, CSS, and JavaScript, demonstrating that modern web technologies can be used to build attractive educational platforms without requiring complex software infrastructure. Users can navigate through lessons, interact with educational materials, complete quizzes, and participate in learning challenges. The project therefore meets its primary objective of promoting health education through digital technology. Although some advanced functionalities such as database integration and automated user management are not yet implemented, the current version provides a strong foundation for future development, and UBUZIMA HUB has the potential to become an effective platform for improving health literacy and empowering young people to make informed life decisions.

4.3 Recommendations

- Integrate a database for permanent data storage and develop an administrator dashboard.
- Allow online user registration with secure authentication.
- Store quiz scores and learning progress and award digital badges and certificates.
- Introduce discussion forums, video-based learning, and health chat support.
- Expand and regularly update the learning content.
- Develop Android and iOS mobile applications.
- Introduce multilingual support (English, Kinyarwanda, French).
- Collaborate with health institutions for content validation.
- **lecturer** continue to provide feedback checkpoints during the semester, not only at final submission; Dedicate more sessions to JavaScript in incoming years.
- **To school:** Partner with relevant institutions for student projects, for example, a health-focused project like mine would benefit greatly from content validation by the Ministry of Health or RBC. Such partnerships would make our work more credible and impactful.

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APPENDICES

Appendix A: GitHub Repository and Live Website

Project Name: UBUZIMA HUB

GitHub Repository: https://github.com/JC-221/UBUZIMA

Live Website: https://jc-221.github.io/UBUZIMA/

Appendix B: User Manual

How to use UBUZIMA HUB:

- Open the website in a web browser (https://jc-221.github.io/UBUZIMA/).
- Explore the Home page to learn about the platform.
- Register or log in to your account (where applicable).
- Navigate to the Learn page and select a health topic.
- Read the lessons and educational resources.
- Complete the quiz for the selected topic.
- Participate in available challenges.
- Contact the administrator through the Contact page if assistance is needed.